Difference-Focusing Fusion Decision Method: An Ensemble Learning Framework and Its Application in Improving Deep Learning Sea–Land Segmentation for Waterline Extraction in Synthetic Aperture Radar Imagery

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***Abstract*— Waterline extraction from synthetic aperture radar (SAR) images can be transformed into a sea–land segmentation task. However, two aspects of deep learning sea–land segmentation have been ignored in the literature: 1) deep learning models are commonly built using whole images rather than focusing on their sea–land transition parts and 2) a higher resolution input may not render a better segmentation result under the constraint of a fixed-size receptive field. Our investigation on the aspects indicates that focusing the modeling process on the sea–land transition parts can benefit the waterline extraction, and the highest resolution may not be the best choice for all pixels. We proposed masked soft intersection over union (MSIoU) loss and the difference-focusing fusion decision (DFFD) ensemble learning method. MSIoU loss incorporates the mask of the transition parts to focus the modeling process on the transition parts. The DFFD ensemble learning method imitates manual labeling and can avoid selecting the resolution of input images. The DFFD ensemble model's member models segment an image's sea and land areas at different resolutions. Then, its fusion model further recognizes the pixels where the member models inconsistently predict sea–land types. We applied the DFFD ensemble model to 10-m-resolution SAR images of the test set. Compared to the traditional single-resolution model, the DFFD ensemble model with MSIoU loss achieved a 10.32–12.58-m accuracy in waterline extraction with a 2.08–2.78-m error reduction in the study area. Moreover, the DFFD framework is independent of data and model choice and can be readily modified for other segmentation tasks.**

***Index Terms*— Deep learning, ensemble learning, framework, segmentation, synthetic aperture radar (SAR), waterline.**

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I. Introduction

GLOBAL ocean coastal zones have been becoming the most frequent and intense area of human activities. In turn, increasing human activities and natural phenomena exacerbate the variation of coastal zones under climate change [1], [2], [3]. The waterline is the instantaneous boundary between sea and land. It plays a vital role in monitoring elements of coastal zones, e.g., coastlines and tidal flat elevations [4], [5], [6], [7], [8]. Remote sensing technology is an essential approach to observing waterlines. It has the advantage of much wider coverage than traditional in situ surveys.

Unlike infrared and optical sensors in remote sensing technology [9], [10], [11], [12], the synthetic aperture radar (SAR) is an active microwave sensor. It can image coastal zones all day all weather [13], [14], [15], [16], [17], [18]. However, SAR imagery suffers from intense speckle noise because of its complex imaging mechanism [13], [14], [19], [20], [21]. Moreover, coastal zones usually have varying terrain [22], [23], [24]. The above makes the automatic and accurate extraction of waterlines from SAR images challenging.

Traditional semiautomatic and automatic waterline extraction methods of remote sensing images mainly fall into two categories: edge detection and threshold segmentation. The edge detection methods directly extract the waterlines from images [8], [25], [26], [27], [28], [29], [30], [31],

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[32], [33], [34], [35], [36]. They may generate discontinuous results when extracting complex waterlines. The segmentation methods segment the sea and land parts of images first [37], [38], [39], [40], [41], [42], [43], [44]. Then, they extract the waterlines from the segmentation binary maps. However, they may encounter challenges in determining the optimal threshold in images of significant terrain variations. Thus, both methods require human-involved postprocessing to extract more accurate waterlines.

In recent years, artificial intelligence techniques, such as deep neural networks (DNNs), have shown great potential in various data mining tasks in the field of satellite oceanography [11], [12], [18], [23], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54]. The application of DNNs has made remarkable progress in sea–land segmentation and subsequent waterline extraction [55], [56], [57], [58], [59], [60], [61], [62]. However, most of the progress focuses on optical images, while more research needs to be done for SAR images. U-Net is a DNN designed explicitly for image segmentation [63]. It is named for its elegant U-shaped network architecture [47], [49], [63], [64], [65]. It classifies the pixels in an image according to their neighborhoods. The neighborhoods are called receptive fields. DeepUNet is a representative sea–land segmentation DNN [57]. As a variant of U-Net, DeepUNet introduces Resblocks [66] to improve the feature extraction ability [57]. DeepUNet performs well in sea–land segmentation in optical images [57], [58], [67].

Samples are fundamental for deep learning technology. However, the following two essential and inspiring aspects are not considered in the current research on deep learning applications in sea–land segmentation.

1) In the sea–land segmentation task, the samples come from three nonoverlapped regions: inner sea, inner land, and the areas between them. We called the areas the sea–land transition bands. The samples from the sea–land transition bands are usually much less than those from the inner sea or inner land regions. However, the sea–land transition bands are concerned with sea–land segmentation and subsequent waterline extraction. We should focus on the sea–land transition bands to avoid the effect of imbalanced samples. The sea–land transition bands around the waterlines have sea and land parts that are similar in size. Thus, using only samples from the sea–land transition bands can also prevent the possible imbalance between sea pixels and land pixels. Both imbalances may negatively affect the model performance in waterline extraction.
2) When manually labeling the sea and land areas, we alternately zoom in and out of the remote sensing images on a computer screen and recognize the sea and land areas at different resolutions to obtain accurate labels. We can explain the labeling procedure below: high resolution tells us more about local details. Complementarily, low resolution lets us get more context information when we can only see the image partially due to a limited field of view. For example, we can only see an oversized image partially on a computer screen. Analogically, when a model recognizes a pixel, the model's receptive field is the model's "screen." Thus, the highest resolution may not always be the best choice for all pixels under the constraint of a fixed-size receptive field.

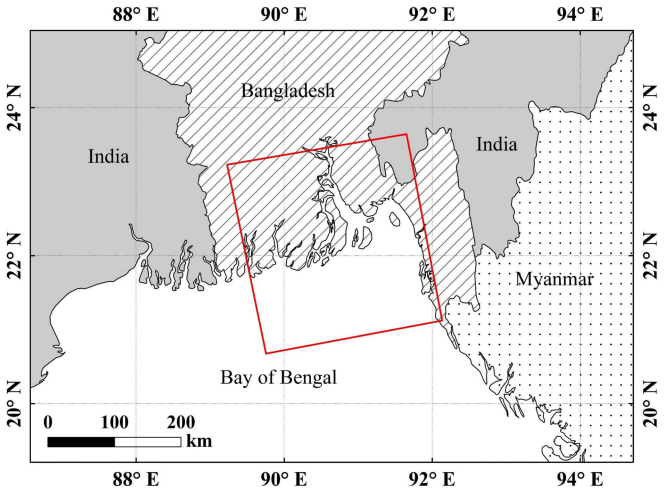


Fig. 1. Study area (enclosed by the red box). The boundaries are from Natural Earth Data [68].

With the investigation of the two aspects, we proposed a loss function—masked soft intersection over union (MSIoU) loss—and an ensemble learning method—the difference-focusing fusion decision (DFFD). MSIoU loss is a variant of soft intersection over union (Soft IoU) loss, incorporating the mask of the sea–land transition bands. MSIoU loss focuses the modeling process on the transition bands. The DFFD ensemble model imitates the manual-labeling procedure. It comprises several member models and a fusion model. The member models segment an image's sea and land areas at different resolutions. Then, the fusion model fuses the pixel types predicted by the member models to recognize the types of the pixels where the member models inconsistently predict sea–land types.

We organize the remainder of this article as follows. Sections II and III introduce the study area, SAR data, and sample sets. Section IV describes DeepUNet and the loss functions (including MSIoU loss) to build the single-resolution models in the study. Section IV also gives the metrics to evaluate sea–land segmentation performance. Section V uses the single-resolution models to investigate the effect of focusing the modeling process on the sea–land transition bands and the effect of image resolution. Section VI proposes the DFFD ensemble learning method. Section VII gives a summary.

II. Study Area and Data

The Bay of Bengal (see Fig. 1) is the largest bay in the world, where the complex environmental and climatic conditions frequently change the coast [69]. Our study area (enclosed by the red box in Fig. 1) lies in the Bay of Bengal, including the delta of the Ganges River.

Sentinel-1A is a C-band SAR satellite launched by the European Space Agency (ESA) Copernicus program in 2014. The interferometric wide (IW) swath mode is one of the primary working modes of the SAR. The IW mode has 10-m

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TABLE I
ACQUISITION TIMES [IN COORDINATED UNIVERSAL TIME (UTC)] OF THE SAR IMAGES AND THEIR USAGES IN THE STUDY

Year-month-day	Hour:minute	Usage
2016-04-29	12:04	Test set
2016-10-14	12:04	Test set
2017-05-06	12:04	Training, validation, and fusion sets
2017-11-02	12:04	Training, validation, and fusion sets
2018-04-19	12:04	Training, validation, and fusion sets
2018-10-28	12:04	Training, validation, and fusion sets
2019-04-14	12:04	Test set
2019-10-11	12:04	Test set

pixel spacing and 250-km swath width and is suitable for coastal monitoring applications. For simplicity, we will refer to pixel spacing as resolution.

We collected two IW SAR images in vertical–vertical polarization each year from 2016 to 2019. One image is in April or May, while the other is in October or November. Thus, there are eight SAR images, and Table I gives the acquisition times of the SAR images.

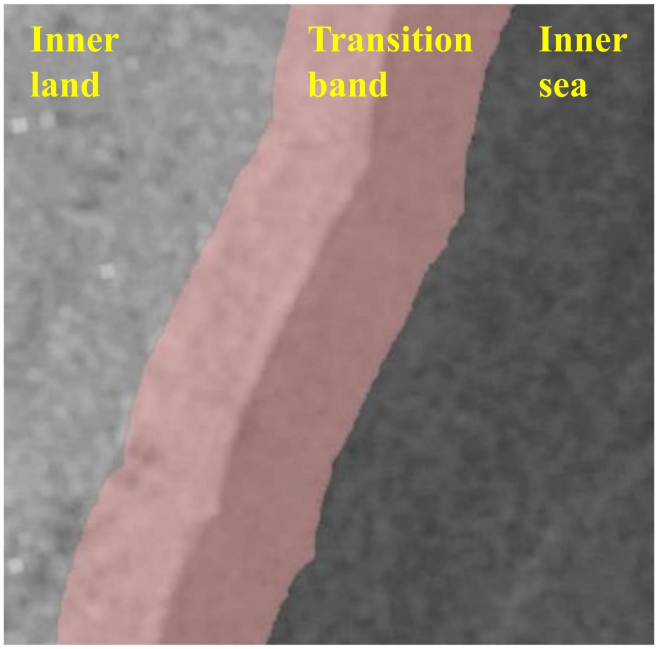


Fig. 2. Example (10-m resolution) illustrates the sea–land transition band. The light red area is the sea–land transition band. The background is the SAR image of the example.

III. SAMPLE SETS

A. Processing SAR Data

We processed the SAR images with radiometric correction, geometric correction, and decibel conversion using the ESA Sentinel Application Platform (SNAP). Next, we visually recognized the sea and land areas of the SAR images and manually labeled the pixel-wise sea–land maps. The maps are binary images where zero and one represent "sea" and "land," respectively. For simplicity, we call them the true sea–land maps.

We compared the 10-, 20-, and 40-m-resolution models to investigate the effect of image resolution on the sea–land segmentation and subsequent waterline extraction (see Section V-B). Thus, we reduced the 10-m resolution of the original SAR images and their sea–land maps to 20- and 40-m resolutions.

Then, we applied the image morphology operation of dilatation to generate the sea–land transition bands centered at the waterlines in the manually labeled sea–land maps. The bands have a bandwidth of 121 pixels. That means the bandwidth is 1.21, 2.42, and 4.84 km for 10-, 20-, and 40-m resolutions, respectively. Fig. 2 gives an example of a band in the 10-m-resolution case.

B. Training, Validation, Fusion, and Test Sets Focusing on the Sea–Land Transition Bands

We used the four SAR images in 2017 and 2018 to generate training, validation, and fusion sets. The fusion set is used to build the fusion model of the DFFD ensemble model in Section VI-B. We cut the original and lower resolution SAR images and sea–land maps into image patches of 252 × 252 pixels and map patches of 68 × 68 pixels at their corresponding resolutions. At the same time, we kept the different-resolution patches concentric and the 40-m-resolution map patches nonoverlapped. The difference in size between image and map patches is because the convolution layers of all models in this study worked in a "valid" manner. (If working in a "valid" manner, the convolution layer will not use padding operation to maintain its output size the same as its input size.) Each pair of an image patch and a map patch at the same resolution is a sample at the resolution. We divided the 10-m-resolution samples whose centers are within the 10-m-resolution sea–land transition bands into the training, validation, and fusion sets. The sample size ratio among the sets is ∼16:4:5. As the different-resolution samples are concentric, we generated the corresponding 20- and 40-m-resolution training, validation, and fusion sets by the centers of samples in the 10-m-resolution sets. Thus, the samples sizes of the 20- and 40-m-resolution sets are the same to those of the 10-m-resolution sets.

We used the four SAR images in 2016 and 2019 to generate the test set. The test set contains samples of concentric 10-, 20-, and 40-m-resolution image patches and 10-m-resolution map patches. The centers of the patches are within the 10-m-resolution sea–land transition bands. Unlike in the training, validation, and fusion sets, we let the 10-m-resolution map patches be nonoverlapped in the test set. Therefore, the sample size of the test set is larger than the other sets. When investigating the effect of image resolution on the sea–land segmentation and subsequent waterline extraction, we upsampled the outputs of the lower resolution models to a 10-m resolution. Then, we compared all the models within the 10-m-resolution sea–land transition bands on the same test set.

Table II gives the sample sizes of the training, validation, fusion, and test sets generated above.

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TABLE II
SAMPLE SIZES OF THE TRAINING, VALIDATION, FUSION, AND TEST SETS

	Training set	Validation set	Fusion set	Test set
Sample size	3311	827	1035	77877

C. Training and Validation Sets Not Focusing on the Sea–Land Transition Bands

We also generated the nonfocal 10-m-resolution training and validation sets of samples from the four SAR images in 2017 and 2018 to compare the modeling ways of focusing and not focusing on the sea–land transition bands (see Section V-A). Unlike the training and validation sets in Section III-B, the centers of samples in these sets are randomly distributed in the whole SAR images rather than being limited within the images' sea–land transition bands. We let the sample sizes of these sets be the same as the training and validation sets in Section III-B.

IV. NETWORK ARCHITECTURE AND LOSS FUNCTIONS OF SINGLE-RESOLUTION MODELS AND PERFORMANCE METRICS

A. DeepUNet

DeepUNet is a network architecture developed for the sea–land segmentation task. It has shown excellent performance in recent literature. Therefore, we adopted it to build the study's single-resolution models whose inputs are single-resolution images. We used the models to investigate the effect of focusing the modeling process on the sea–land transition bands in Section V-A and the effect of image resolution in Section V-B.

DeepUNet is a variant of U-Net where the convolution layers are replaced with Resblocks to enhance the feature extraction ability (see Fig. 3). The input of the DeepUNet model receives an image, and the DeepUNet model outputs the score map of the image. The size of the score map is smaller than that of the input image, as the model implements convolutions in a "valid" manner rather than the "same" manner in this study. The value at a pixel in the score map is between zero and one. It can be considered as the probability of the pixel belonging to the land. Finally, the binarization with a threshold of 0.5 transfers the score map into a sea–land map where one and zero mean land and sea, respectively.

B. Loss Functions

Intersection over union (IoU) is a widely used metric to evaluate segmentation performance

$$\mathrm{IoU} = \frac{|X \cap Y|}{|X \cup Y|}. \tag{1}$$

X is the true object area, while Y is the model-predicted object area. IoU can measure the difference between two areas and thus evaluate segmentation performance. If expressed in pixels, (1) becomes

$$\mathrm{IoU} = \frac{\sum_{\mathrm{pixels}} Y_{\mathrm{true}} Y_{\mathrm{pred}}}{\sum_{\mathrm{pixels}} Y_{\mathrm{true}} + \sum_{\mathrm{pixels}} Y_{\mathrm{pred}} - \sum_{\mathrm{pixels}} Y_{\mathrm{true}} Y_{\mathrm{pred}}} \tag{2}$$

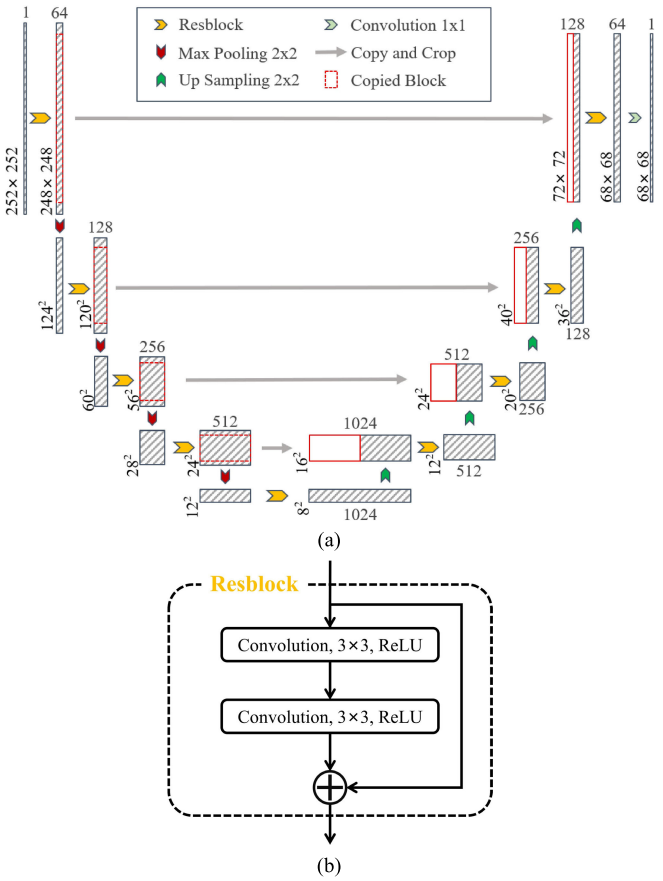


Fig. 3. Architecture of DeepUNet and Resblock. (a) Architecture of DeepUNet. The dark red arrow represents the 2 × 2 max pooling operation, the dark green arrow represents the 2 × 2 upsampling operation, the light green arrow represents the 1 × 1 convolution, the gray arrow represents the cropping and copying operation, and the yellow arrow represents Resblock. (b) Architecture of Resblock. Resblock contains two cascaded 3 × 3 convolutions (with ReLU activation function) and an addition operation. "+" represents the addition operation.

where Y_{true} and Y_{pred} are the values at the pixels in the true and model-predicted sea–land maps. IoU ranges from zero to one, with a higher value indicating better segmentation performance.

The IoU's opposite number cannot be used as a loss function, as the binarization operation from a model-predicted score map to a model-predicted sea–land map is nondifferentiable. Soft IoU loss is a relaxed version of IoU's opposite number, which is differentiable

$$L_{\mathrm{SoftIoU}} = -\frac{\sum_{\mathrm{pixels}} Y_{\mathrm{true}} y_{\mathrm{pred}}}{\sum_{\mathrm{pixels}} Y_{\mathrm{true}} + \sum_{\mathrm{pixels}} y_{\mathrm{pred}} - \sum_{\mathrm{pixels}} Y_{\mathrm{true}} y_{\mathrm{pred}}} \tag{3}$$

where y_{pred} is the value at pixels in the model-predicted score map.

In this study, we proposed MSIoU loss to focus the modeling process on the sea–land transition bands

$$L_{\mathrm{MSIoU}} = -\frac{\sum_{\mathrm{pixels}} w_{\mathrm{band}} Y_{\mathrm{true}} y_{\mathrm{pred}}}{\sum_{\mathrm{pixels}} w_{\mathrm{band}} Y_{\mathrm{true}} + \sum_{\mathrm{pixels}} w_{\mathrm{band}} y_{\mathrm{pred}} - \sum_{\mathrm{pixels}} w_{\mathrm{band}} Y_{\mathrm{true}} y_{\mathrm{pred}}} \tag{4}$$

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TABLE III

IOUS OF THE MODELS IN THE SEA–LAND TRANSITION BANDS ON THE TEST SET AND ITS PARTS. PARTS 1–4 OF THE TEST SET ARE GENERATED FROM THE SAR IMAGES ON APRIL 29, 2016, OCTOBER 14, 2016, APRIL 14, 2019, AND OCTOBER11, 2019, RESPECTIVELY. THE INTEGERS ARE THE SAMPLE SIZES OF THE TEST SET AND ITS PARTS. THE BOLD NUMBERS ARE THE HIGHEST IOUS, AND THE UNDERLINED NUMBERS ARE THE SECOND HIGHEST IOUS

Models	IoU (%)				
	Whole test set (77877)	Part one (25168)	Part two (25507)	Part three (13682)	Part four (13520)
Nonfocal 10-meter-reolution model	92.72	89.56	94.84	94.65	93.00
10-meter-resolution model	95.61	95.01	96.42	<u>96.20</u>	94.56
20-meter-resolution model	95.76	95.08	96.36	95.78	**95.89**
40-meter-resolution model	94.87	94.68	94.84	94.87	95.31
DFFD ensemble model	**96.18**	**95.75**	**96.75**	**96.21**	<u>95.88</u>
Bagging ensemble model	95.65	95.15	96.29	95.79	95.22
Random forest ensemble model	96.04	95.63	96.64	96.07	95.61
XGBoost ensemble model	<u>96.10</u>	<u>95.70</u>	<u>96.69</u>	96.14	95.67

where w_{band} represents the mask of the sea–land transition bands. If a pixel is within the sea–land transition bands, the value of w_{band} is one; otherwise, it is zero.

C. Performance Metrics

Besides IoU in Section IV-B, we also evaluate the models in this study based on the waterlines in their predicted sea–land maps.

First, we mosaiced the model-predicted sea–land map patches in the test set into four sea–land maps. The mosaiced sea–land maps are corresponding to the four SAR images that generate the test set (see Section III-B). The sea area is the maximal water-body-connected region in the study area. Thus, we extracted the edges of the maximal water-body-connected parts of the mosaiced sea–land maps. We considered the extracted edges as the model-predicted waterlines of the four SAR images. Similarly, we extracted the waterlines of the four SAR images from the manually labeled sea–land maps and considered them as the true waterlines of the four SAR images.

Second, according to the study area, we transformed the coordinate system of the model-predicted and true waterlines into the Universal Transverse Mercator projection coordinate system (WGS_1984_UTM_Zone_46N). Next, we calculated the shortest distance from each point on the model-predicted waterlines to the true waterlines in the projection coordinate system. Then, we calculated the root mean square of the distances (RMSD). We used the waterline RMSD as a metric to evaluate the models' performance.

V. INVESTIGATING TWO ASPECTS

A. Investigating the Effect of Focusing the Modeling Process on the Sea–Land Transition Bands

We built two 10-m-resolution sea–land segmentation models. The first model is built traditionally using the nonfocal 10-m-resolution training and validation sets (see Section III-C) and Soft IoU loss (3). We call the first model the nonfocal 10-m-resolution model. The second model is built using the 10-m-resolution training and validation sets focusing on the sea–land transition bands (see Section III-B) and MSIoU loss (4). We call the second model the 10-m-resolution model. Using the Adam optimization algorithm, we trained the models on their training sets [70]. Then, we reserved the model parameters with the minimum loss on their validation sets to avoid overfitting. We set the batch size to 64, the maximum iteration number to 500, and the initial learning rate to 0.01.

After training, we evaluated the models' performance in the sea–land transition bands on the test set (see Section III-B) using IoU and the waterline RMSD.

The first two rows of Table III show that the 10-m-resolution model achieved higher IoUs than the nonfocal 10-m-resolution model on the test set and all its parts corresponding to the four SAR images in 2016 and 2019. Moreover, the first two rows of Table IV show that the waterlines predicted by the 10-m-resolution model also have smaller RSMDs than the nonfocal 10-m-resolution model in all four SAR images. The results indicate that focusing the modeling process on the sea–land transition bands can help improve model performance.

B. Investigating the Effect of Image Resolution

We used the 20- and 40-m-resolution training and validation sets (see Section III-B) to build the 20- and 40-m-resolution models, respectively. We also used MSIoU loss as the loss function. The optimization algorithm, batch size, maximum iteration number, and initial learning rate are the same as in Section V-A (Adam, 64, 500, and 0.01). After training, we applied the 20- and 40-m-resolution models to the test set and upsampled their outputs to 10-m resolution. Then, we compared the performance of the 20- and 40-m-resolution

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TABLE IV

RSMDs OF THE WATERLINES EXTRACTED FROM THE SEA–LAND MAPS PREDICTED BY THE MODELS ON THE TEST SET. THE BOLD NUMBERS ARE THE SMALLEST RMSD, AND THE UNDERLINED NUMBERS ARE THE SECOND SMALLEST RMSD

Models	RSMD (m)			
	Image one 2016-04-29	Image two 2016-10-14	Image three 2019-04-14	Image four 2019-10-11
Nonfocal 10-meter-reolution model	14.98	13.27	12.40	14.71
10-meter-resolution model	13.72	10.54	10.88	13.35
20-meter-resolution model	12.88	11.37	10.93	**11.26**
40-meter-resolution model	19.64	18.43	16.36	15.36
DFFD ensemble model	**12.58**	**10.49**	**10.32**	12.07
Bagging ensemble model	16.15	14.89	14.94	16.61
Random foreast ensemble model	13.49	11.62	11.65	13.88
XGBoost ensemble model	13.15	11.16	11.09	13.35

models and the 10-m-resolution model built in Section V-A in the sea–land transition bands on the test set.

In rows 2–4 of Table III, we can see that the 20-m-resolution model achieved an overall higher IoU than the 10- and 40-m-resolution models on the test set. The 20-m-resolution model is better than the 10- and 40-m-resolution models on the first and fourth parts of the test. The 10-m resolution model is better on the second and third parts.

To show the difference in IoU among the models more intuitively, we used the kernel density estimation method [71] to estimate the models' IoU probability density functions (pdfs) of samples in the test set

$$\hat{f}(\text{IoU}) = \frac{1}{Nh}\sum_{n=1}^{N} K\left(\frac{\text{IoU} - \text{IoU}_n}{h}\right) \tag{5}$$

where $\hat{f}(\text{IoU})$ is the estimated pdf. IoU_n is the IoU of the nth sample, and N is the sample size. $K(x)$ is the kernel function. Here we used a Gaussian kernel

$$K(x) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{1}{2}x^2\right). \tag{6}$$

The parameter h in (5) is the bandwidth of the kernel function and is determined as follows:

$$h = 1.06\sigma N^{-0.2} \tag{7}$$

where σ is the standard deviation of IoUs of samples. Fig. 4 gives the estimated IoU pdfs of the 10-, 20-, and 40-m-resolution models. The 20-m-resolution model has more sample IoUs distributed close to one than the 10- and 40-m-resolution models.

We also compared the waterlines predicted by the 10-, 20-, and 40-m-resolution models in the four SAR images of the test set. The 20-m-resolution model achieved the smallest waterline RSMDs among the three single-resolution models in the first and fourth SAR images (see rows 2–4 of Table IV). It also achieved the second smallest waterline RSMD among the three single-resolution models in the second and third SAR images (see rows 2–4 of Table IV).

Thus, the 20-m-resolution model outperforms the 10- and 40-m-resolution models on the test set. The results indicate that a higher resolution sometimes leads to a better effect, but not always.

The first three columns of Fig. 5 give examples of misclassified areas (denoted in red) of the 10-, 20-, and 40-m-resolution models. We calculated the IoUs of the models in the areas enclosed by two cyan boxes in each example. In the first example [see Fig. 5(a)–(c)], the best single-resolution models for the left and right areas are the 10- and 20-m-resolution models, respectively. In the second example [see Fig. 5(e)–(g)], they are the 20- and 40-m-resolution models. In the third example [see Fig. 5(i)–(k)], they are the 10- and 20-m-resolution models.

Therefore, even in the same image, the higher resolution does not always lead to a better result—the proper resolution can differ in different places.

VI. DFFD ENSEMBLE LEARNING METHOD

A. Idea of the DFFD Ensemble Learning Method

Section V-B suggests that classifying the pixels in different parts of an image could require different resolutions. Thus, it naturally brings forward the problem of selecting a proper resolution. From the view of a single model, we can alleviate the problem by deepening the multiscale network of the model. However, a larger model size will dramatically increase the computation cost and training difficulty. In contrast, an ensemble model can perform well by combining its small-size member models. Thus, we explore alleviating the problem from the view of an ensemble model.

The results in Section V-B are consistent with the intuition: when we visually classify a pixel in an image on a fixed-size computer screen, we zoom in and out of the image and recognize the pixel at different resolutions (the first

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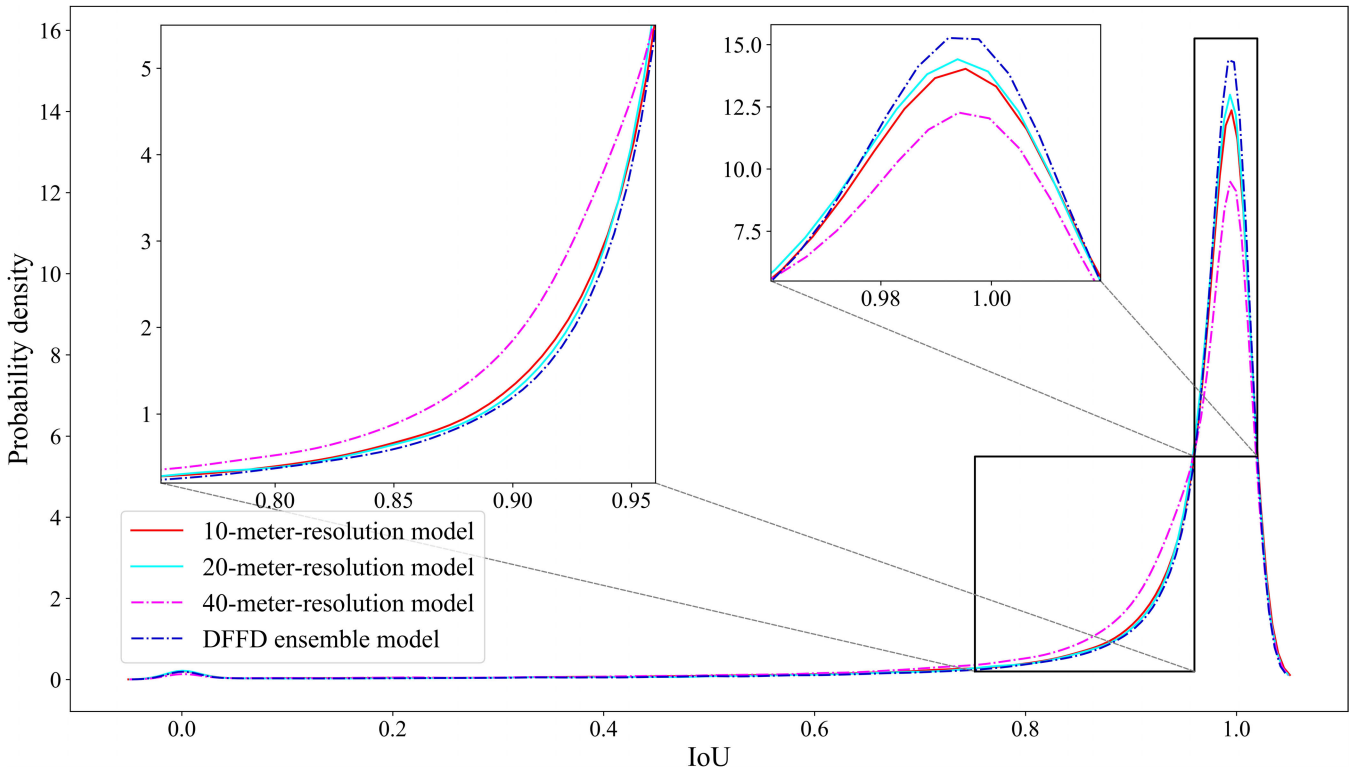


Fig. 4. IoU pdfs of the 10-, 20-, and 40-m-resolution models and the DFFD ensemble model (in the sea–land transition bands) on the test set.

stage). Then, we comprehensively judge the pixel's type (the second stage). A lower resolution can provide more context information about the pixel as the screen can show a larger neighborhood of the pixel. Complementarily, a higher resolution can provide more information on local details.

Analogically, when a model recognizes a pixel, the model's receptive field is the model's "screen." The analogy inspired us to propose an ensemble learning method for the segmentation task: the DFFD method. The DFFD ensemble model consists of several member models and a fusion model (see Fig. 6). The member models are individual segmentation models. The DFFD ensemble learning method imitates the image-labeling procedure to predict pixel types with two stages.

In the first stage, the method downsamples the original image and uses the member models to recognize the pixels at different resolutions. The first stage of the method is analogous to the first stage of the image-labeling procedure, in which we recognize the pixels at various resolutions.

If all the member models classify a pixel as the same type, then the type of the pixel is decided. Otherwise, the method implements the second stage, where the fusion model will further recognize the pixel based on the pixel types predicted by the member models. The second stage imitates the second stage of the image-labeling procedure, in which we make a comprehensive decision on the pixel's type based on the type of judgments made at different resolutions. As seen above, the fusion model focuses on recognizing the pixels with inconsistent-type predictions made by the member models. These pixels are usually hard-to-recognize ones.

The details of building and applying the DFFD ensemble model will be given in Sections VI-B and VI-C, respectively.

B. Building the DFFD Ensemble Model

For illustration, we take the 10-, 20-, and 40-m-resolution DNN models built in Section V as the member models of the DFFD ensemble model.

To demonstrate the effectiveness of the DFFD framework, we used a very simple network with attention mechanism as the fusion model (see Fig. 7). In the fusion model, the cascade of a fully connected layer, rectified linear unit (ReLU) activation function, and softmax function generates and normalizes three nonnegative weights from three input scores. Next, the input scores are multiplied by the weights and summed. The weighted sum is the score output by the fusion model. As the input scores range from zero to one and the sum of the nonnegative weights is one, the output score also ranges from zero to one. This light fusion model only has one fully connected layer to train, which has 12 parameters.

We applied the member models to the fusion set (see Section III-B). Then, according to the pixel types predicted by the member models, we divided the pixels in the fusion set into consistent and inconsistent ones. A consistent pixel is a pixel that the member models classify as the same type, while an inconsistent pixel is one that the member models classify as different types. The fusion model receives the member-model-predicted scores at inconsistent pixels. Then, the fusion model predicts new scores at inconsistent pixels.

The objective of optimizing the fusion model is to improve the performance of the whole ensemble model on both consistent and inconsistent pixels rather than only on inconsistent pixels. Therefore, MSIoU loss function for

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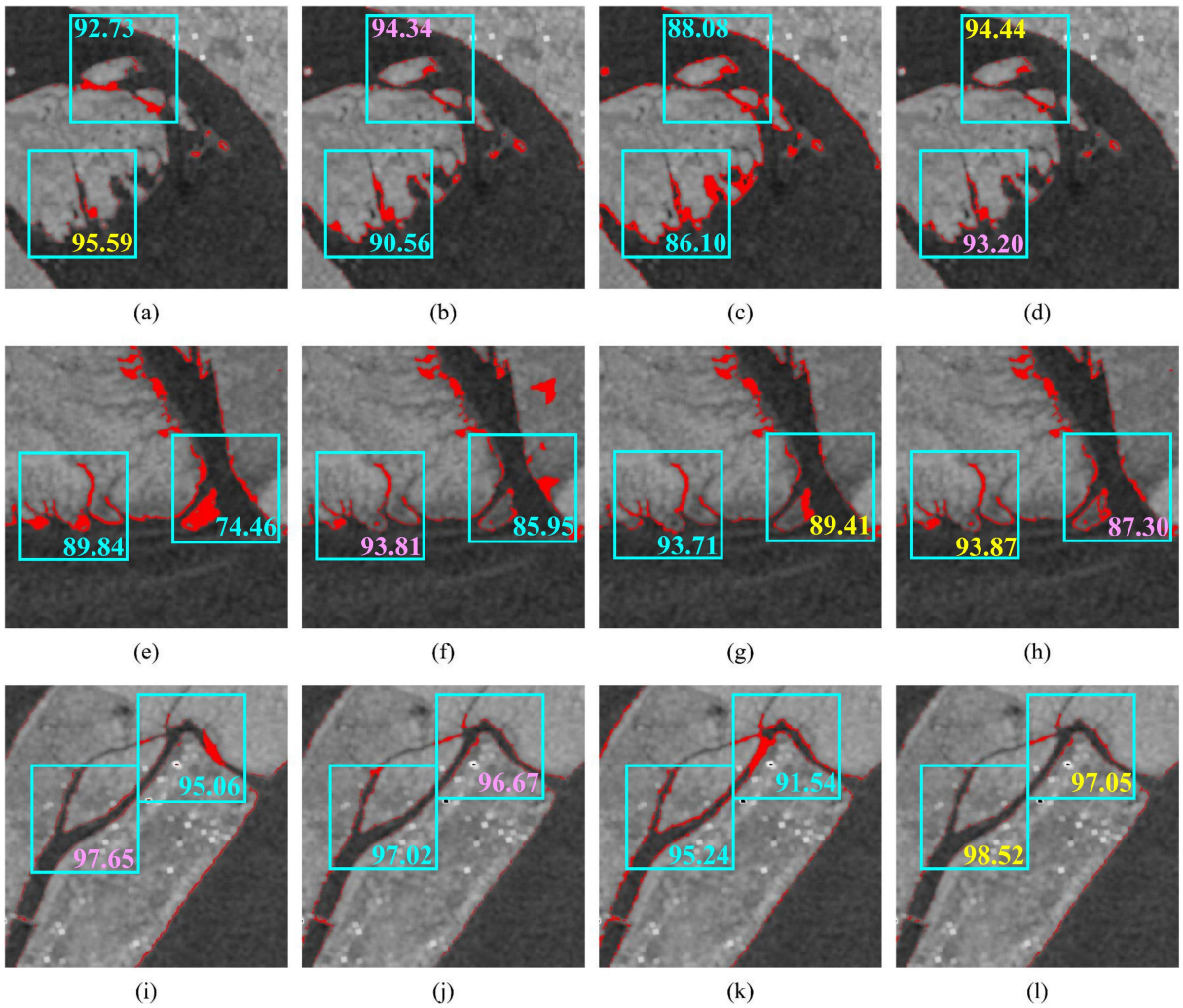


Fig. 5. Three examples illustrate the areas misclassified by the single-resolution and DFFD ensemble models on the test set. (a), (e), and (i) 10-m-resolution model; (b), (f), and (j) 20-m-resolution model; (c), (g), and (k) 40-m-resolution model; and (d), (h), and (l) DFFD ensemble model. The red areas are the misclassified areas. We also calculate the IoUs (yellow numbers expressed in percentage) in the cyan boxes for quantitative comparison between the models. The backgrounds are the SAR images of the examples.

optimizing the fusion model should be

$$L_{\text{MSIoU_fusion}} = -\frac{A + \sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}}{\left(B + \sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} + \sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{pred}} - \sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}\right)} \tag{8}$$

where

$$A = \sum_{\text{pixels}\notin P} w_{\text{band}} y_{\text{true}} y_{\text{pred}} \tag{9}$$

$$B = \sum_{\text{pixels}\notin P} w_{\text{band}} y_{\text{true}} + \sum_{\text{pixels}\notin P} w_{\text{band}} y_{\text{pred}} - \sum_{\text{pixels}\notin P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}. \tag{10}$$

P represents the set of inconsistent pixels. A and B are constants in training the fusion model. The member models determine A and B regardless of the fusion model. Thus, they can be considered as prior knowledge acquired by the member models.

Note that directly applying MSIoU loss (4) to inconsistent pixels will generate

$$L_{\text{MSIoU_inconsistent}} = -\frac{\sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}}{\left(\sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} + \sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{pred}} - \sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}\right)}. \tag{11}$$

Equations (8) and (11) can be equivalently transformed into

$$L_{\text{MSIoU_fusion}} = -\frac{\frac{A}{\sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}} + 1}{\frac{B}{\sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}} + \frac{1}{L_{\text{MSIoU_inconsistent}}}} \tag{12}$$

and

$$L_{\text{MSIoU_inconsistent}} = -\frac{\sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}}{\sum_{\text{pixels}\in P} w_{\text{band}} - \sum_{\text{pixels}\in P} w_{\text{band}} (1 - y_{\text{true}})(1 - y_{\text{pred}})}. \tag{13}$$

The loss functions $L_{\text{MSIoU_fusion}}$ and $L_{\text{MSIoU_inconsistent}}$ reach their global minimums at the same time in the ideal case

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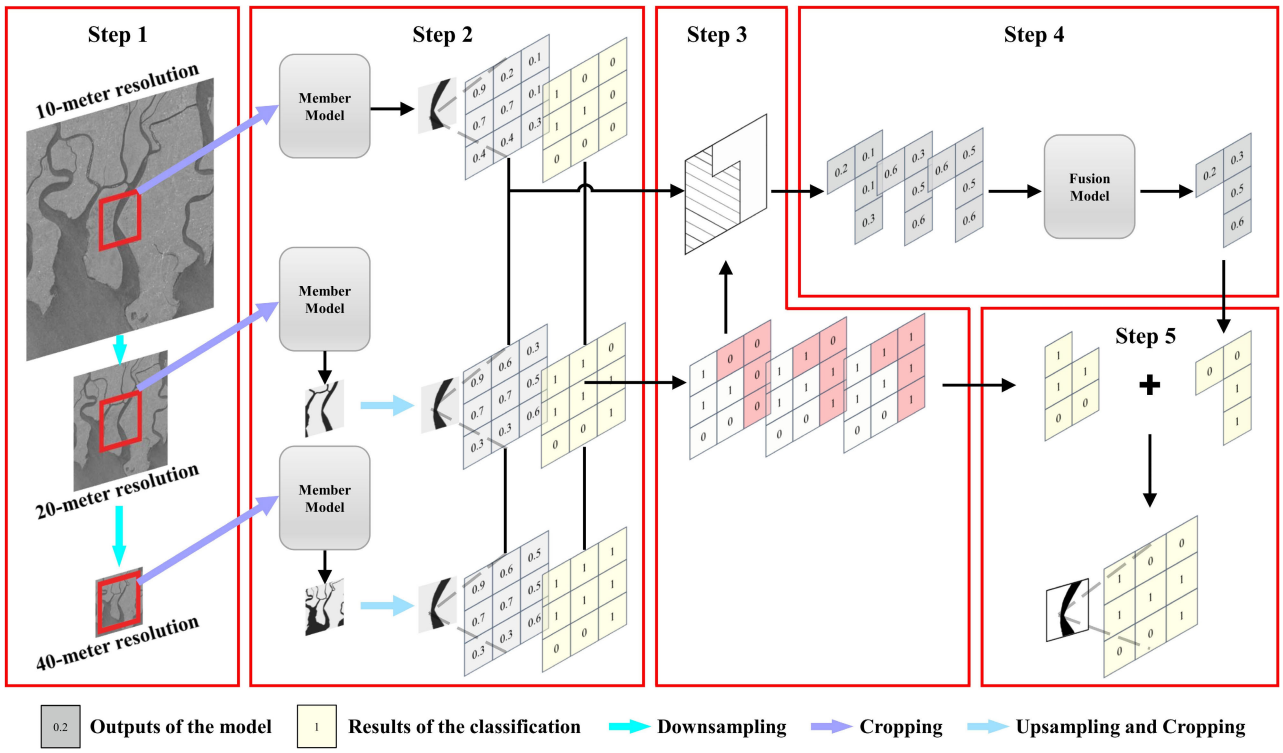


Fig. 6. Schematic of the DFFD ensemble model. The black arrow represents the data flow in the method without operations. The bold plus sign in Step 5 represents the mosaic operation.

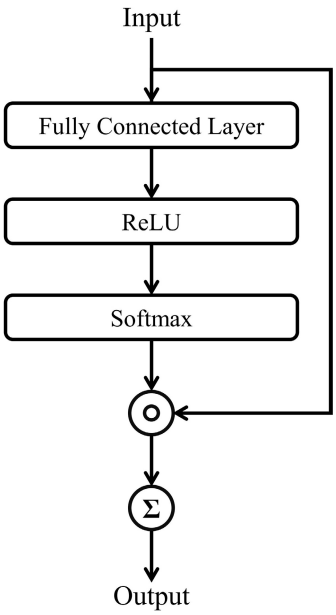


Fig. 7. Architecture of the fusion model in the study. "∘" and "Σ" represent the Hadamard product and the sum operation, respectively.

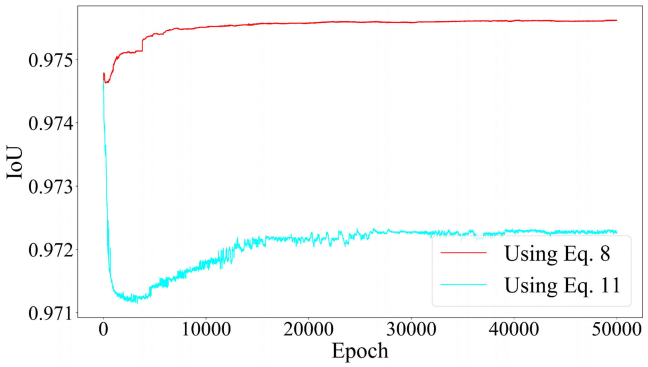


Fig. 8. IoU variations on the fusion-model validation set when using different loss functions to training the fusion model. The red curve represents using $L_{\text{MSIoU_fusion}}$ (8). The cyan curve represents using $L_{\text{MSIoU_inconsistent}}$ (11).

that $y_{\text{pred}} = y_{\text{true}}$ at all pixels belong to P. However, comparing (8) and (11), we can see that $L_{\text{MSIoU_fusion}}$ is different from $L_{\text{MSIoU_inconsistent}}$ due to the additional terms A and B. Equation (13) indicates that $L_{\text{MSIoU_inconsistent}}$ and $\sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}$ can be considered independent due to the denominator on the right side of (13). Thus, a decreasing $L_{\text{MSIoU_inconsistent}}$ cannot always lead to a decreasing $L_{\text{MSIoU_fusion}}$, as $L_{\text{MSIoU_inconsistent}}$ can decrease under constant $L_{\text{MSIoU_fusion}}$. In other words, different values of $L_{\text{MSIoU_inconsistent}}$ can correspond to the same value of $L_{\text{MSIoU_fusion}}$ due to $\sum_{\text{pixels}\in P} w_{\text{band}} y_{\text{true}} y_{\text{pred}}$ on the right side of (12). Therefore, if we use $L_{\text{MSIoU_inconsistent}}$ as the loss function to optimize the fusion model in the training process, the optimization direction could deviate from the objective of optimizing the fusion model. Even the optimization could fall into a local optimum corresponding to a larger value of $L_{\text{MSIoU_fusion}}$ and, subsequently, a worse performance of the whole ensemble model.

To present the difference in performance between $L_{\text{MSIoU_fusion}}$ and $L_{\text{MSIoU_inconsistent}}$, we used them as the loss function to train the fusion model, respectively. To avoid overfitting, we divided the inconsistent pixels of samples in the fusion set into two parts: one for training the fusion model (called the fusion-model training set) and the other for

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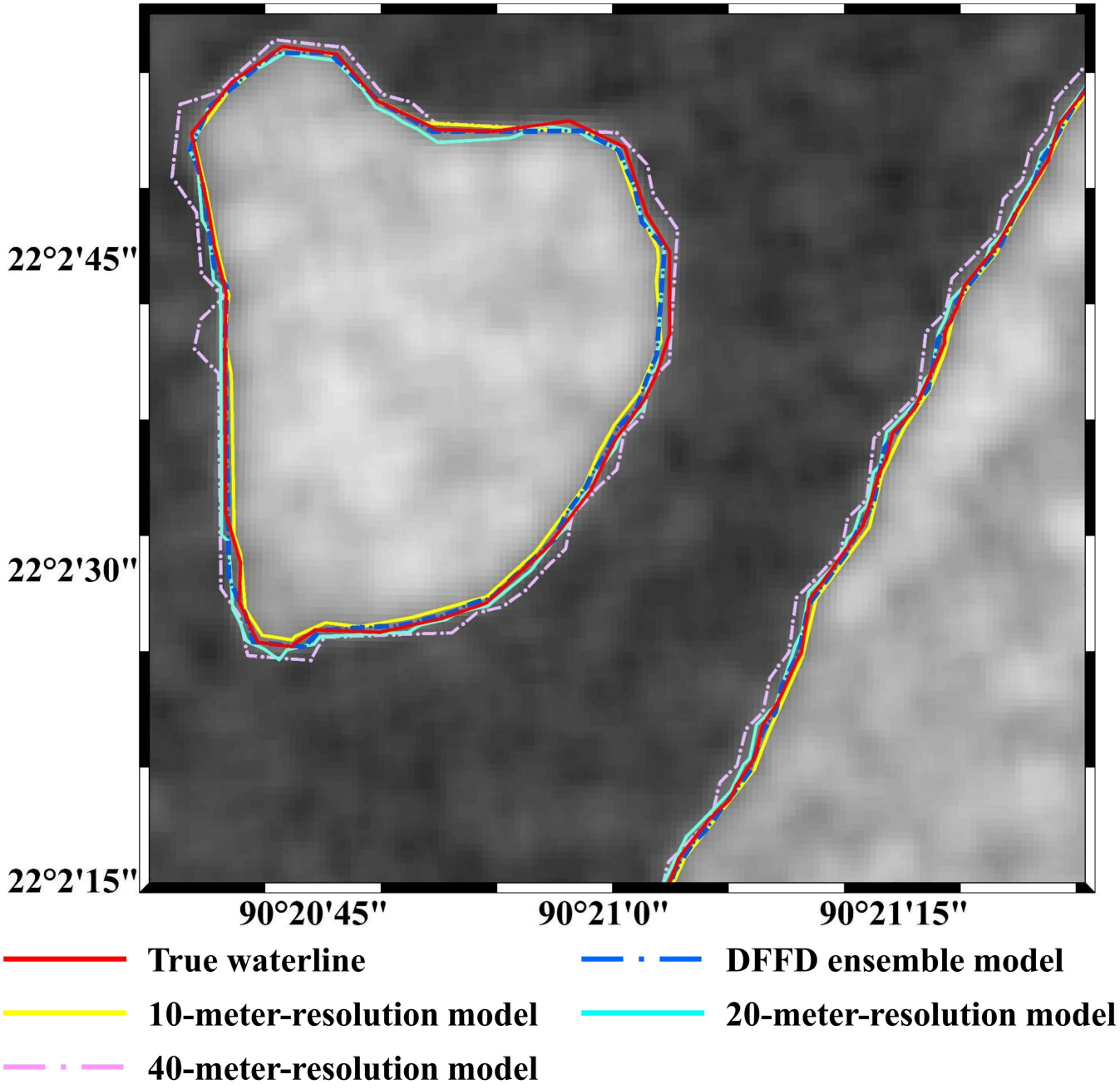


Fig. 9. Example illustrates the waterlines extracted from the sea–land maps predicted by the single-resolution and DFFD ensemble models on the test set. The red solid lines represent the manually labeled waterlines. The blue dashed-dot lines represent the waterlines predicted by the DFFD ensemble model. The yellow solid lines represent the waterlines predicted by the 10-m-resolution model. The cyan solid lines represent the waterlines predicted by the 20-m-resolution model. The pink dashed-dot lines represent the waterlines predicted by the 40-m-resolution model. The background is the SAR image of the example.

validating the fusion model (called the fusion-model validation set). We trained the fusion model using the Adam optimization algorithm. We set the batch size to 512, the initial learning rate to 0.01, and the ratio between the sample sizes of the two sets to ∼1:1.

Fig. 8 shows the IoU variations on the fusion-model validation set during the training of the fusion model under the two loss function settings. From Fig. 8, we can see that when using $L_{\text{MSIoU_inconsistent}}$ as the loss function, the IoU even decreases during the initial phase. Afterward, the IoU increases but converges to a lower value. In contrast, when using $L_{\text{MSIoU_fusion}}$ as the loss function, the IoU can converge faster and more smoothly to a higher value. In other words, using $L_{\text{MSIoU_fusion}}$ as the loss function can take fewer training iterations to obtain good-performing model parameters.

Therefore, the DFFD method uses $L_{\text{MSIoU_fusion}}$ as the loss function to build the fusion model.

C. Applying the DFFD Ensemble Model

Take the 10-m-resolution SAR image in the top-left corner of Fig. 6 to illustrate the prediction procedure of the DFFD ensemble model.

Step 1: Downsample the original 10-m-resolution SAR image to 20- and 40-m resolutions. Crop the 10-, 20-, and

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40-m-resolution images to obtain three concentric images (enclosed by the red boxes in Fig. 6) containing a concerned area.

Step 2: Input the three concentric images to the member models of the corresponding resolutions and obtain three member-model-predicted score maps. Upsample the score maps predicted by the lower resolution member models to 10-m resolution and crop them to the exact size of the score map output by the 10-m-resolution member model. Binarize the three score maps of the same size to the sea–land maps with a score threshold of 0.5.

Step 3: Compare the member-model-predicted sea–land maps (generated at the end of Step 2) pixel by pixel at the 10-m resolution and divide the pixels into consistent and inconsistent ones.

Step 4: Input the member-model-predicted scores of the inconsistent pixels to the fusion model and get their fusion-model-predicted scores. Binarize the fusion-model-predicted scores to the sea–land types with a score threshold of 0.5.

Step 5: Mosaic the sea–land types predicted by the member models at the consistent pixels and those predicted by the fusion model at the inconsistent pixels to get the ensemble-model-predicted sea–land map (i.e., the DFFD-predicted sea–land map).

D. Results of the DFFD Ensemble Model

We chose the optimal parameters of the fusion model in Section VI-B with the minimum value of $L_{\text{MSIoU_fusion}}$ on the fusion-model validation set. Then, we applied the DFFD ensemble model to the test set. As shown in Fig. 5, the DFFD ensemble model improved the sea–land segmentation IoUs over the 10-, 20-, and 40-m-resolution models in most areas enclosed by the cyan boxes. The example in Fig. 9 shows that the DFFD-predicted waterlines (blue dashed-dot lines) are closer to the true waterlines (red solid lines) than those (other lines) predicted by the 10-, 20-, and 40-m-resolution models.

We calculated the IoU, IoU pdf, and waterline RMSD of the DFFD ensemble model in the sea–land transition bands on the test set. Table III shows that the DFFD ensemble model achieved the highest IoU on the test set and, specifically, is the best on the first, second, and third parts of the test set and the second best on the fourth part. Compared with the 10-, 20- and 40-m-resolution models, the DFFD ensemble model also has IoUs of more samples distributed close to one (see Fig. 4). Table IV shows that the DFFD ensemble model achieved the smallest waterline RMSD in the first, second, and third images of the test set and the second smallest waterline RMSD in the fourth image.

Compared with the traditional model (the nonfocal 10-m-resolution model), the IoU of the DFFD ensemble model increased by 1.56%–6.19% (see Table III). At the same time, its waterline RMSD decreased by 2.08–2.78 m (see Table IV).

E. Comparison With Other Ensemble Learning Methods

For comparison, we also used Bagging [72], random forest [73], and XGBoost [74] ensemble learning methods to fuse the pixel-type predictions made by the 10-, 20-, and 40-m-resolution models. We built the Bagging, random forest, and XGBoost models using the fusion set and fed them the scores predicted by the 10-, 20-, and 40-m-resolution models. Then, we applied them to the test set. The last three rows of Tables III and IV give their IoUs (in the sea–land transition bands) and waterline RMSD. We can see that the DFFD ensemble model outperforms them on the test set and all its four parts in both IoU and waterline RMSD.

VII. Summary

We investigated two crucial but easily overlooked aspects in the current research on deep learning sea–land segmentation for waterline extraction.

1) Deep learning models are frequently built using whole SAR images without specific emphasis on the sea–land transition bands.
2) As inspired by the manual-labeling procedure, a higher resolution input may not yield a better result under the constraint of a fixed-size receptive field.

Our results indicate that focusing the modeling process on the transition bands helps improve the model performance in the transition bands, thus benefiting the waterline extraction. Additionally, higher resolution leads to better performance sometimes, but not always. Moreover, the proper resolution can differ in different places, even in the same image.

We proposed MSIoU loss and the DFFD ensemble learning method. MSIoU loss incorporates the mask of the transition bands to focus the modeling process on the transition bands. The DFFD ensemble learning method imitates manual labeling and can avoid selecting the resolution of input images.

The DFFD ensemble model with MSIoU loss outperforms the single-resolution models (10-, 20-, and 40-m resolutions) on the test set. It achieves an accuracy varying from 95.75% to 96.75% in IoU in the transition bands and from 10.32 to 12.58 m in waterline RMSD on the four parts of the test set. Compared with the traditional nonfocal single-resolution model (10-m resolution and not focusing the modeling process on the transition bands), it made a 1.56%–6.19% improvement in IoU in the transition bands and a 2.08–2.78-m reduction in waterline RMSD.

The main framework of the DFFD ensemble learning method is independent of input data, choice of member and fusion models, and output classification categories. The types of member models can be different in the DFFD ensemble model. We could improve the DFFD ensemble model under the framework in various ways. For example, we could use more advanced member and fusion models, develop other loss functions to incorporate the prior knowledge acquired by the member models, and feed more information to the fusion model. Moreover, the method could also alleviate the issue of input selection in other factors, such as polarization and bands of input remote sensing images, by constructing the member models to process images with different selections in the factors. Consequently, the method holds substantial promise for deployment across a spectrum of image segmentation tasks.

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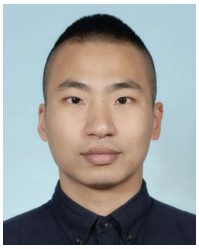
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